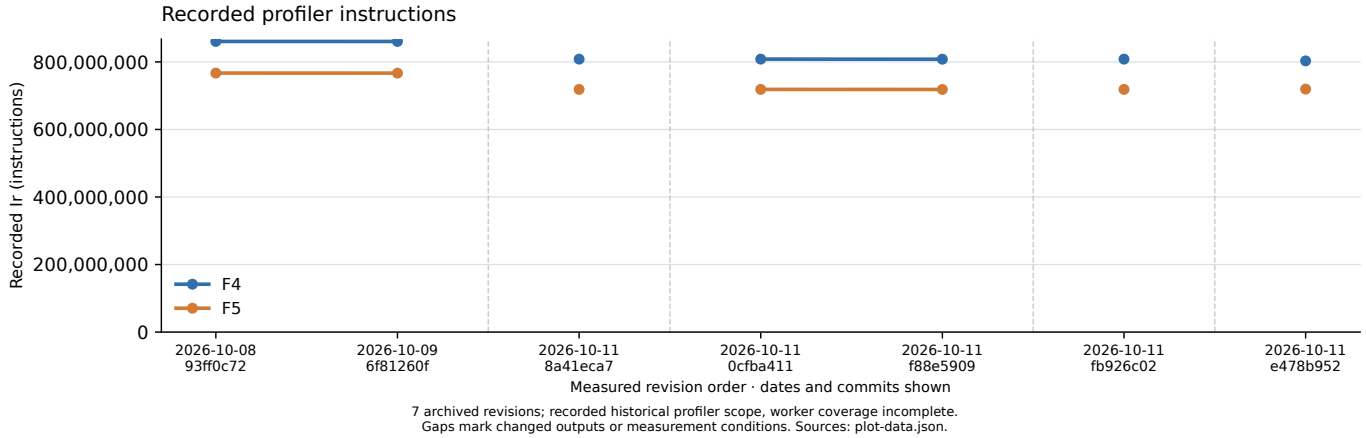
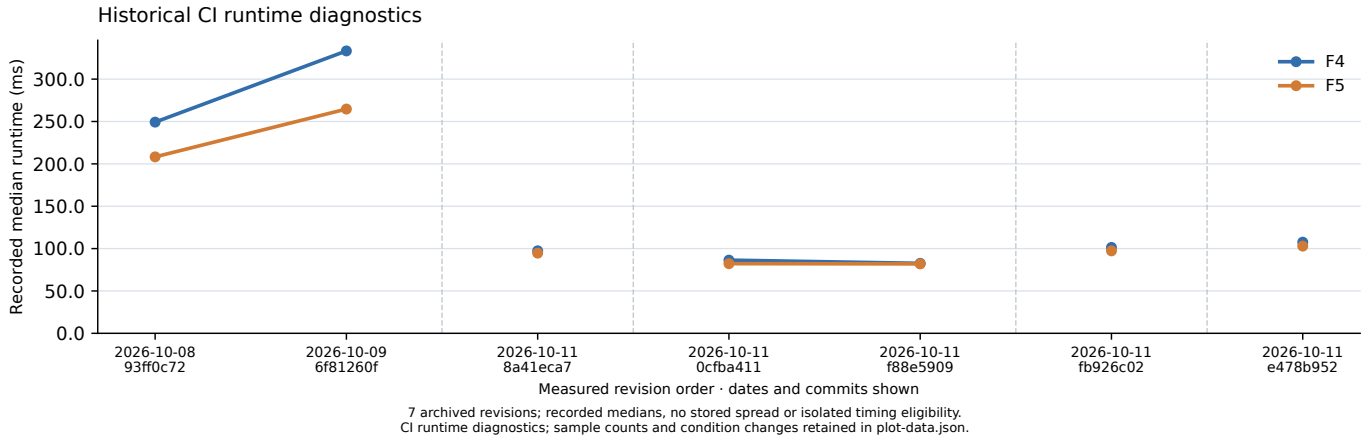
Boolean matrix step: F4 and F5

One shared K_1 / 2^23 m=2 system, 22 Boolean variables, maximum degree 4. Matrix construction, reduction and readback; F5 includes its criterion.





UTC date / revision	F4 Ir	F5 Ir	F4 ms	F5 ms
2026-10-08 / 93ff0c72	860,633,889	766,940,469	249.241	208.250
2026-10-09 / 6f81260f	860,555,567	766,776,087	333.279	264.687
2026-10-11 / 8a41eca7	808,263,705	718,576,235	97.380	94.658
2026-10-11 / 0cfba411	808,263,705	718,578,523	86.390	82.214
2026-10-11 / f88e5909	807,989,935	718,391,055	82.597	81.948
2026-10-11 / fb926c02	808,263,705	718,576,189	101.379	97.323
2026-10-11 / e478b952	803,017,657	719,516,063	107.585	102.927

Historical Python-orchestrated perf.history/v1 snapshots; no new performance run or speedup claim.

Runtime: one recorded median per kernel and revision; sample counts retained per point, no stored spread, no isolated timing eligibility. Runtime is diagnostic.

Profiler Ir is the recorded historical scope. Worker-thread coverage is incomplete for these kernels; it is not a complete solver operation count.

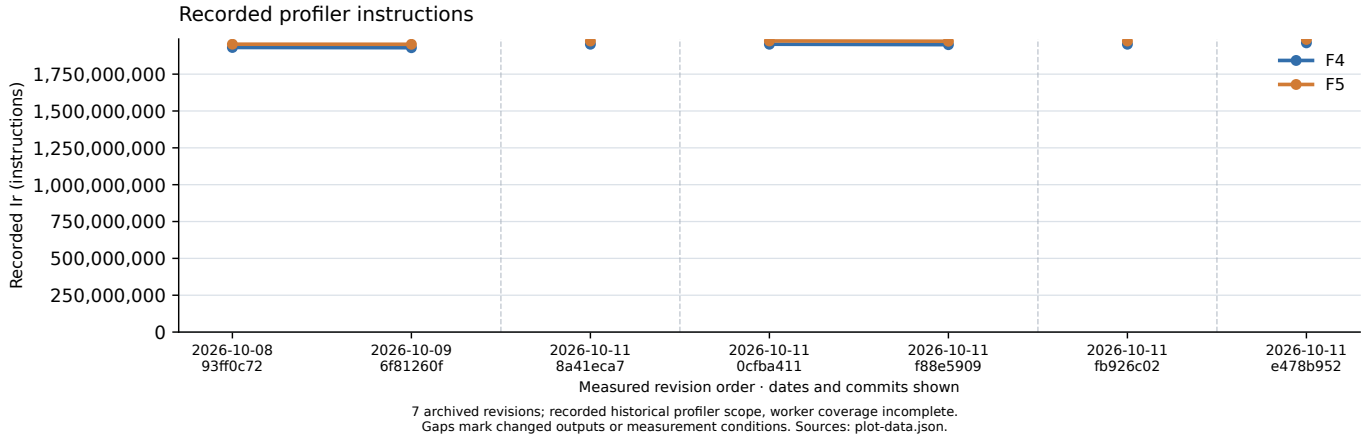
Lines stop at missing values, changed output fingerprints, or changed host/compiler/harness/profiler. Different algorithm fingerprints are expected and are not equated.

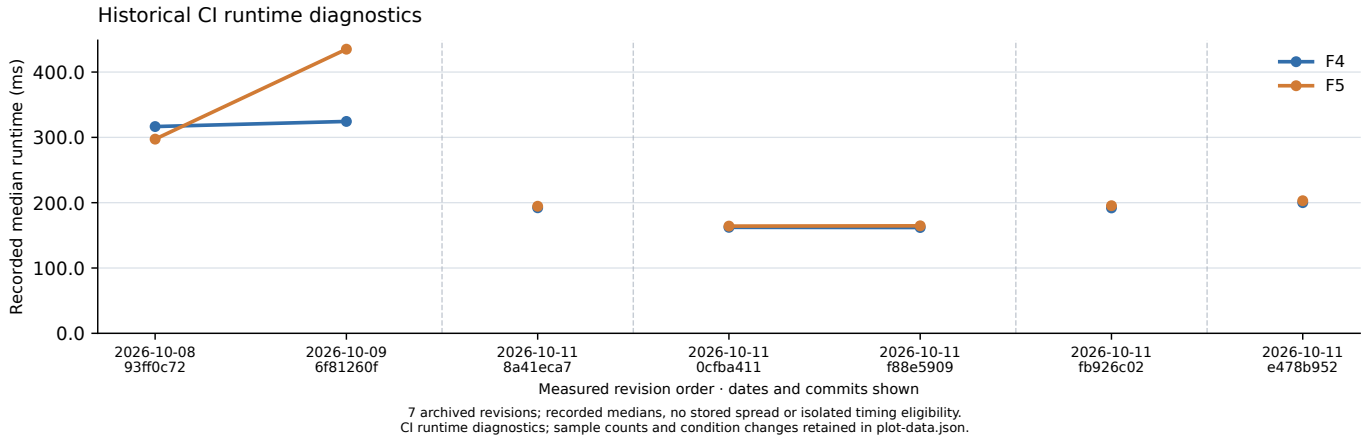
X positions are measured revision order, labelled with UTC commit date and revision; spacing is not elapsed time.

Evidence: 7 imported perf.history/v1 snapshots; historical_sources in plot-data.json retains raw filenames, SHA-256 hashes, hosts, harnesses and sample counts. Source data and provenance: plot-data.json and REPORT.md. These plots do not establish an end-to-end ECDLP gain.

Boolean solver calls: F4 and F5

Shared K_1 / 2^17 m=2 workload, 16 Boolean variables, four planted and four random targets, maximum degree 3. The solve_boolean_system entry point, including root readback.





UTC date / revision	F4 Ir	F5 Ir	F4 ms	F5 ms
2026-10-08 / 93ff0c72	1,930,989,200	1,953,108,992	316.570	297.297
2026-10-09 / 6f81260f	1,929,374,570	1,952,598,745	324.403	435.004
2026-10-11 / 8a41eca7	1,953,021,023	1,974,789,730	192.203	194.561
2026-10-11 / 0cfba411	1,953,020,766	1,975,267,516	162.266	164.161
2026-10-11 / f88e5909	1,949,981,977	1,973,219,339	161.982	164.584
2026-10-11 / fb926c02	1,953,020,766	1,974,789,781	191.988	195.374
2026-10-11 / e478b952	1,962,376,386	1,985,422,672	200.058	202.996

Historical Python-orchestrated perf.history/v1 snapshots; no new performance run or speedup claim.

Runtime: one recorded median per kernel and revision; sample counts retained per point, no stored spread, no isolated timing eligibility. Runtime is diagnostic.

Profiler Ir is the recorded historical scope. Worker-thread coverage is incomplete for these kernels; it is not a complete solver operation count.

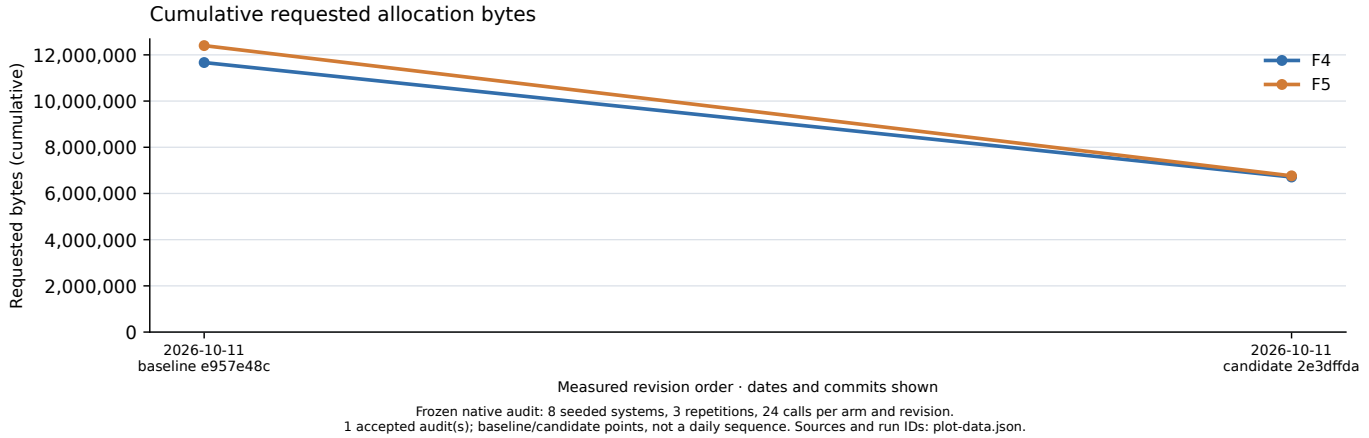
Lines stop at missing values, changed output fingerprints, or changed host/compiler/harness/profiler. Different algorithm fingerprints are expected and are not equated.

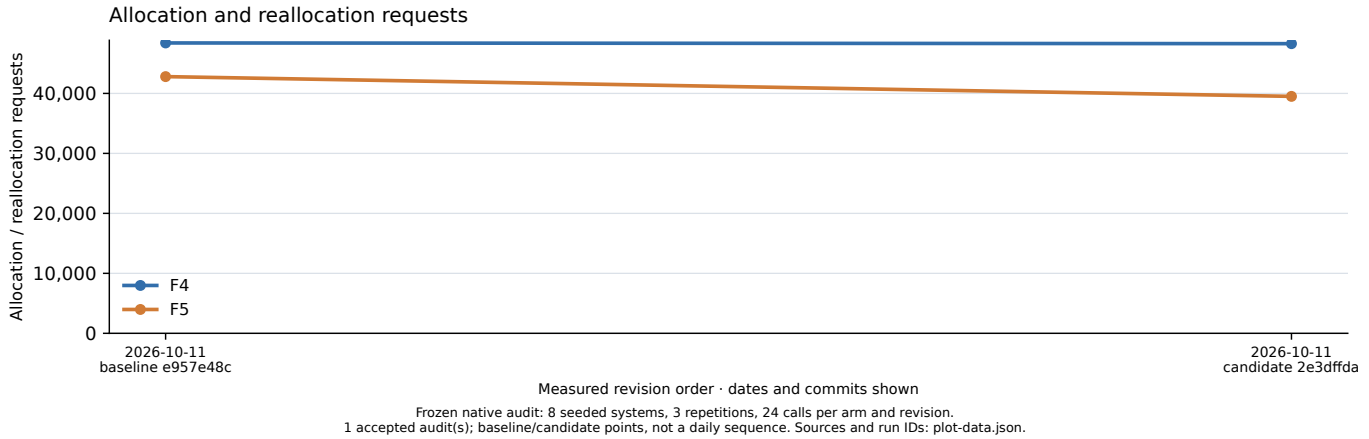
X positions are measured revision order, labelled with UTC commit date and revision; spacing is not elapsed time.

Evidence: 7 imported perf.history/v1 snapshots; historical_sources in plot-data.json retains raw filenames, SHA-256 hashes, hosts, harnesses and sample counts. Source data and provenance: plot-data.json and REPORT.md. These plots do not establish an end-to-end ECDLP gain.

Complete Boolean calls: allocation history

Matching frozen Boolean allocation suite: 8 seeded systems, 3 repetitions, 24 calls per arm and revision. Polynomial dimensions, degrees and seeds are retained in the source audit protocol.





UTC date / revision	F4 bytes	F5 bytes	F4 requests	F5 requests
2026-10-11 / e957e48c	11,667,696	12,400,920	48,414	42,801
2026-10-11 / 2e3dffda	6,712,608	6,762,696	48,294	39,507

Native allocation audit. Original baseline/candidate rows, accepted comparison, exact outputs and accounting, shared inputs and all totals revalidated by the Rust plot-data tool.

One baseline and one candidate point per accepted audit, not a daily sequence. A/A and replication pairs are controls, not additional algorithm revisions.

Bytes are cumulative requested traffic, not peak memory. No timing is measured by this allocation series.

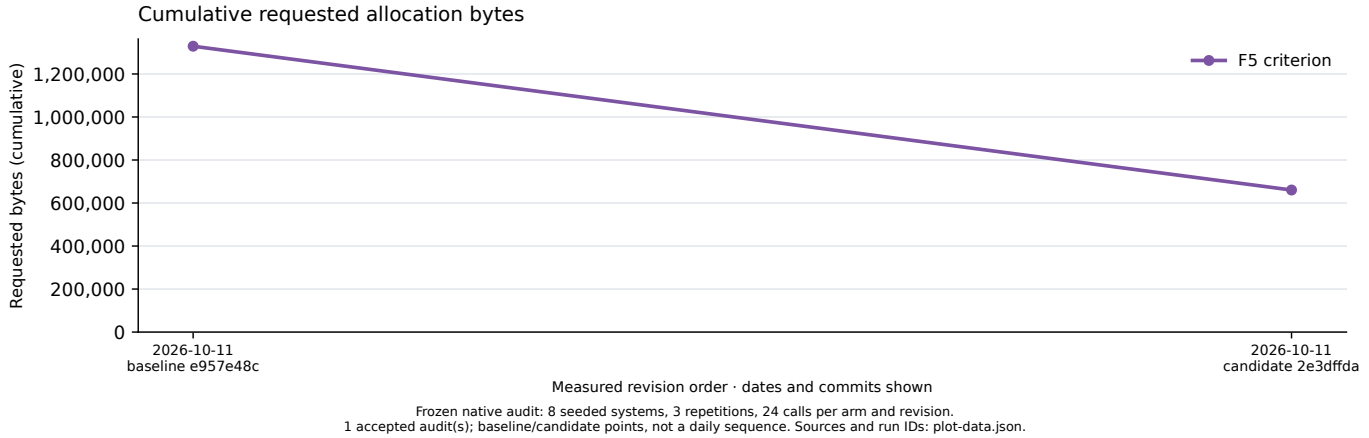
Criterion construction is a separate part of F5 and is never compared as a complete solver.

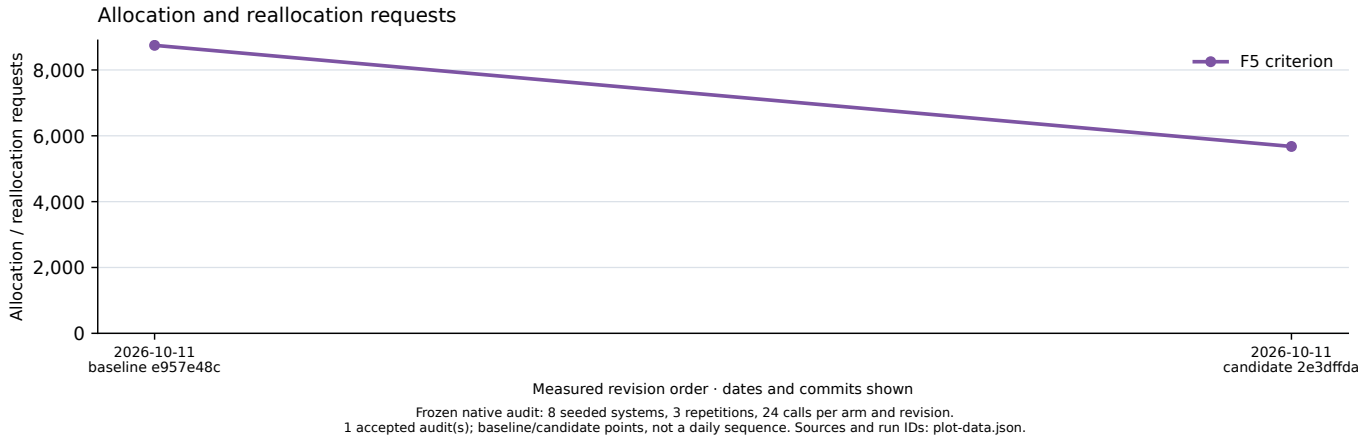
Audit revisions come from their frozen records. They are not inserted as measurements of historical CI workloads; the workloads and metrics remain separate.

Evidence: allocation_sources in plot-data.json; frozen native comparison and baseline/candidate JSONL records in research/f4_f5_row_build_20261011/records/. Source data and provenance: plot-data.json and REPORT.md. These plots do not establish an end-to-end ECDLP gain.

F5 criterion: allocation history

Matching frozen Boolean allocation suite: 8 seeded systems, 3 repetitions, 24 calls per arm and revision. Polynomial dimensions, degrees and seeds are retained in the source audit protocol.





UTC date / revision	F5 criterion bytes	F5 criterion requests
2026-10-11 / e957e48c	1,329,024	8,748
2026-10-11 / 2e3dffda	660,480	5,676

Native allocation audit. Original baseline/candidate rows, accepted comparison, exact outputs and accounting, shared inputs and all totals revalidated by the Rust plot-data tool.

One baseline and one candidate point per accepted audit, not a daily sequence. A/A and replication pairs are controls, not additional algorithm revisions.

Bytes are cumulative requested traffic, not peak memory. No timing is measured by this allocation series.

Criterion construction is a separate part of F5 and is never compared as a complete solver.

Audit revisions come from their frozen records. They are not inserted as measurements of historical CI workloads; the workloads and metrics remain separate.

Evidence: allocation_sources in plot-data.json; frozen native comparison and baseline/candidate JSONL records in research/f4_f5_row_build_20261011/records/. Source data and provenance: plot-data.json and REPORT.md. These plots do not establish an end-to-end ECDLP gain.